

The color of Earth's sky, 4.4 billion years to today

What the sky looked like from the ground—straight up and at the horizon, at the equator and the poles, at noon and at dusk—at thirteen moments in Earth's history, reconstructed with a spectral radiative-transfer model.

Prepared with Claude (Anthropic) · September 2026

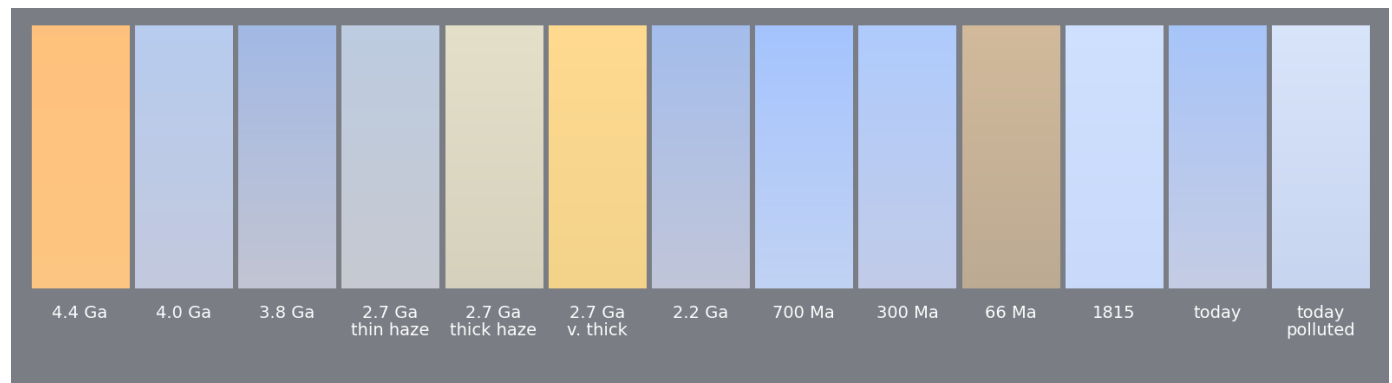


Figure 1. Mid-latitude noon sky at thirteen epochs, zenith (top) to horizon (bottom), oldest at left. Brightness shown relative to today's clean sky.

Five things the sky did

1. Under the 30-bar CO_2 atmosphere of the early Hadean the sky was a shadow-less peach-white glow with no visible Sun.
2. During the hazy Neoproterozoic the whole dome was cream to apricot, with almost no difference between zenith and horizon.
3. Before the Great Oxidation Event there was no ozone, so every twilight ended in a cream zenith. The first ozone layer turns that zenith pale blue; the deep blue dusk needs a near-modern column.
4. Snowball Earth had the bluest sky in Earth's history; the K–Pg impact winter had no sunsets at all.
5. The zenith-to-horizon color gradient is the single best diagnostic of an atmosphere's state: largest in clean air, gone under haze or soot, reversed under volcanic sulfate.

Has this been done before?

Not as such. A search of the Valency corpus (arXiv, PubMed, Earth-ArXiv and others) finds the two halves of the problem solved separately but never joined. On one side, atmospheric-optics work computes modern sky color rigorously: Lynch and Mazuk (2005) derive CIE chromaticities of the sky, Sun and distant objects from MODTRAN spectra; Peyvandi et al. (2016) simulate 5.6 million daylight spectra with SBDART to map how aerosol and ozone shift sky color; Lee and Mollner (2017) show with Monte Carlo simulation that vivid solar-horizon twilight colors need aerosol while the purest antisolar colors occur in a purely molecular atmosphere. On the other side, paleoatmosphere work constrains composition: Catling and Zahnle (2020) for the Archean (O_2 below 10^{-6} of present, CO_2 10–2,500 $\times$ modern, CH_4 10^2 – $10^4\times$ modern), and Arney et al. (2016), Mak et al. (2023) and Goldblatt et al. (2024) for the organic-haze episodes. Arney's "Pale Orange Dot" computes what a hazy Archean Earth looks like *from space*, as an exoplanet spectrum, and that is as close as the literature comes. Nobody has computed the view from the ground, or the horizon-versus-zenith, latitude, and sunset structure. So this report does.

How the colors were computed

For each epoch the atmosphere is built from components with their own vertical profile and spectral optical depth: Rayleigh-scattering gas (N_2 , O_2 , CO_2 , CH_4 , H_2O in their epoch-appropriate partial pressures), ozone (the Chappuis cross section of Serdyuchenko et al. 2014 at 223 K, on a layer whose peak falls from 26 km at the equator to 18 km at the pole), tropospheric aerosol, a stratospheric sulfate layer, a tholin-like fractal organic haze (strongly blue-absorbing, forward-scattering), and impact soot. Sky radiance is integrated numerically

along each line of sight through a spherical-shell atmosphere, so grazing sunlight at twilight is handled properly, with multiple scattering and surface reflection added from a delta-Eddington two-stream solution. The young Sun is treated as a blackbody made fainter and slightly cooler following standard solar evolution: 5,560 K and 70% of today's luminosity at 4.4 Ga, 5,620 K and 75% at 3.8 Ga, 5,660 K and 80% at 2.7 Ga, 5,700 K and 85% at 2.2 Ga, 5,772 K today. The color effect is only ~ 200 K, barely perceptible and far smaller than the atmospheric effects; the brightness effect is what makes the early skies dimmer. Spectra from 380–780 nm are converted to CIE XYZ and shown as sRGB. The model reproduces measured modern values (clear zenith 10,000–20,000 K, horizon 6,000–8,500 K, setting Sun $\sim 1,700$ K, pink antisolar Belt of Venus) before being pointed backward.

The young Sun. The Sun is treated as a blackbody whose effective temperature and luminosity follow standard solar-evolution tracks: 5,560 K and 70% of today's output at 4.4 Ga, rising through 5,620 K (75%) at 3.8 Ga, 5,660 K (80%) at 2.7 Ga and 5,700 K (85%) at 2.2 Ga to 5,772 K today (Table 1). The color effect is small: a 5,560 K Sun is only about 200 K cooler in color temperature than the modern one, a difference that is barely perceptible side by side and is swamped by atmospheric effects ten to a hundred times larger. The brightness effect is not small, and is why the early-epoch skies are rendered dimmer. Fraunhofer lines and the young Sun's ultraviolet excess, which matter for haze photochemistry but not for visible color, are ignored.

The model is provided as the Python source `skymodel.py`; `run_epochs.py` produces the sky colors and `limb_grid.py` the limb and disk colors for the renderings. Epoch atmospheres are summarized in Table 1.

Findings

Horizon versus zenith

In a clean molecular atmosphere the zenith is blue and the horizon white because the horizon path is long enough that blue is scattered back out before reaching the eye and multiple scattering fills in the other colors. The size of that gradient is the clearest single diagnostic of atmospheric state. It is largest in the cleanest atmospheres (Snowball Earth, today's clean air: 8,000–13,000 K of color-temperature difference), shrinks as Rayleigh depth rises (late Hadean), collapses to nothing when a broadband absorber or thick scatterer sits overhead (hazy Archean, impact winter, 30-bar Hadean), and actually *reverses* under a stratospheric sulfate veil or heavy pollution, where the zenith

Table 1: Atmospheres used for each epoch. Gas amounts are partial pressures in bar; AOD is tropospheric aerosol optical depth at 550 nm; haze, sulfate and soot are column optical depths at 550 nm; T_{eff} and $L_{\odot}$ are the assumed solar effective temperature and luminosity relative to today.

Epoch	Gas (bar)	O ₃ (DU)	AOD	Haze	Sulfate	Soot	T_{eff} (K)	$L_{\odot}$
4.4 Ga Hadean	CO ₂ 30, N ₂ 1, H ₂ O 0.3	0	0.30	–	–	–	5560	0.70
4.0 Ga Hadean	N ₂ 1, CO ₂ 0.5	0	0.15	–	–	–	5600	0.72
3.8 Ga Archean	N ₂ 0.8, CO ₂ 0.1, CH ₄ 0.001	0	0.08	–	–	–	5620	0.75
2.7 Ga thin haze	N ₂ 0.8, CO ₂ 0.05, CH ₄ 0.003	0	0.08	0.15	–	–	5660	0.80
2.7 Ga thick haze	N ₂ 0.8, CO ₂ 0.05, CH ₄ 0.005	0	0.08	0.6	–	–	5660	0.80
2.7 Ga very thick haze	N ₂ 0.8, CO ₂ 0.05, CH ₄ 0.01	0	0.08	1.5	–	–	5660	0.80
2.2 Ga post-GOE	N ₂ 0.8, O ₂ 0.002, CO ₂ 0.02	150	0.10	–	–	–	5700	0.85
700 Ma Snowball	N ₂ 0.78, O ₂ 0.02, CO ₂ 0.01 (+dust 0.05)	250	0.02	–	–	–	5750	0.94
300 Ma Carboniferous	N ₂ 0.78, O ₂ 0.33	330	0.15	–	–	–	5765	0.975
66 Ma impact winter	modern air	300	0.15	–	0.5	1.5	5772	0.995
1815 volcanic year	modern air	300	0.10	–	0.4	–	5772	1.00
Modern, clean	N ₂ 0.78, O ₂ 0.21, Ar 0.01	300	0.10	–	–	–	5772	1.00
Modern, polluted	modern air	300	0.60	–	–	–	5772	1.00

Table 2: Summary of sky colors by epoch (mid-latitude noon unless stated).

Epoch	Noon zenith	Horizon	Sunset
4.4 Ga, 30-bar CO ₂	Peach-white, 3,300 K, shadowless	Same as zenith	Whole dome dims to orange; Sun rarely visible
4.0 Ga, 0.5-bar CO ₂	Pale milky blue, 10,500 K	Milk white	Deep red Sun; cream twilight (no ozone)
3.8 Ga, clear Archean	Blue, 13,000 K	White	Modern-like, then yellow-beige twilight zenith
2.7 Ga, thick haze	Cream-tan, 5,600 K	Cream-tan	Vermilion horizon, amber sky, long orange dusk
2.2 Ga, post-GOE	Blue, 13,300 K	White	Pale blue-gray twilight zenith (~7,100 K); deep blue dusk comes later
700 Ma, Snowball	Deep blue, 17,000 K	Blue-white, stays blue	Golden, pale
300 Ma, 33% O ₂	Blue, 13,600 K	White, slightly brighter	Modern-like; smoky more often
66 Ma, impact winter	Dim amber-beige, 4,600 K	Beige	None; dome fades to brown
1815, volcanic year	Milky blue-white, 9,300 K	Slightly bluer than zenith	Salmon Sun, pink-lavender afterglow
Ozone-hole spring	Blue, still	White	Pale blue dusk zenith, ~7,800 K
Today, clean	Blue, 15,000 K	White, 8,400 K	Orange horizon, pink antisolar; twilight zenith ~12,500 K
2100, megaconstellation	Same blue as today	White	Thousands of bright datacenters along the dusk terminator, plus a few hundred communications satellites
θ Columbae	B8:IV	5.00, 221 pc	-5.00 (-5.77 to -4.41)2.24.73100%
ε Canis Majoris (Adara)	B1III	1.50, 124 pc	-4.35 (-4.44 to -4.26)8.44.42100%
β Canis Majoris (Mirzam)	B1III	1.98, 151 pc	-3.58 (-3.72 to -3.45)11.74.35100%
ρ Orionis	K0 III	4.46, 105 pc	-3.16 (-4.24 to -2.40)3.12.54100%
Algol	B7V	2.09, 28 pc	-2.88 (-4.65 to 2.09)2.95.0762%
ζ Leporis	A2IV-V(n)	3.55, 22 pc	-2.54 (-2.63 to -2.43)1.30.85100%
ν Puppis	B7III	3.17, 114 pc	-2.42 (-2.64 to -2.22)8.73.56100%
ζ Sagittarii (Ascella)	A2.5Va	2.60, 27 pc	-1.97 (-2.07 to -1.87)3.31.05100%
Canopus	A9II	-0.62, 95 pc	-1.92 (-2.01 to -1.82)52.23.18100%
Aldebaran	K5III	0.87, 20 pc	-1.50 (-1.53 to -1.47)6.80.3392%
ζ Canis Majoris (Furud)	B2.5 V	3.02, 111 pc	-1.47 (-1.87 to -0.94)14.03.2751%

becomes whiter than the horizon.

Equator, mid-latitudes, poles

Latitude enters through solar elevation and surface albedo. A high Sun lights the sky from above through a short path, giving a brighter but paler zenith (equatorial noon ~9,000 K); a low Sun sends light through more air, so the polar summer zenith is a deeper, darker blue (~19,000 K today, 20,000 K over Snowball ice) while the polar horizon is warmer and yellower. The same geometry makes the poles the place where any overhead absorber is most visible: the hazy Archean sky is cream at the equator but distinctly orange-yellow at the poles, and the impact-winter Sun that is still a pale disk at the equator is gone at 75° latitude. Snow and ice brighten and blue-shift the whole dome by reflecting light back into the atmosphere, which is why polar and Snowball horizons stay blue where a mid-latitude land horizon goes white.

Particulates

Three particle regimes appear in the record and they look different from one another. Non-absorbing sulfate high in the stratosphere (volcanic years) whitens the day sky and produces long, luminous pink and lavender twilights, because the layer is still sunlit after the ground is dark. Absorbing organic haze (Neoproterozoic) or soot (impact winter) removes blue outright, turning the dome tan or amber with almost no directional structure, and dims the surface by factors of

3–20. Boundary-layer pollution or dust brightens and grays the zenith, and buries the setting Sun in murk before it reaches the horizon. Tropospheric aerosol is also what makes sunset color vivid: the model confirms Lee & Mollner’s result that a purely molecular sky yields a pale, yellow-orange horizon while any turbidity deepens it to red.

Sunsets and sunrises through time

The biggest surprise is ozone. Today the sky above a setting Sun grades from orange through pale yellow to a zenith that stays blue well into twilight: about 12,500 K with the Sun 4° down. That persistent blue is the Chappuis band, ozone absorbing yellow and orange along the grazing path. The same modern sky with the ozone removed drops that zenith to cream, about 6,000 K. Before the Great Oxidation Event there was no ozone, so every sunset for the first two billion years ended with the zenith fading through cream and beige to orange. At ~2.2 Ga a 66 DU column, the Cooke et al. (2021) global mean for 1% of present oxygen, turns that cream zenith a pale blue-gray near 7,100 K. The deep twilight blue arrives later, once the column is near modern: the snowball sky, at their 10% oxygen column of 169 DU, holds a dusk zenith near 10,800 K. The same thinning is visible in the ozone-hole years. A polar-spring column of 130 DU leaves the dusk zenith pale blue, near 7,800 K, against 12,500 K today. Beyond the ozone story: Hadean sunsets were not events at the horizon but a slow dimming of the whole glowing dome; hazy Archean sunsets were the longest and most saturated in Earth’s history, vermillion at the horizon and amber overhead; Snowball sunsets were brief and

golden; the sunsets of the impact winter did not exist as color at all; and the reddest, most theatrical skies of the Phanerozoic belong to volcanic years like 1815 and 1883.

The atmosphere from space

To show latitude and altitude at once, these renderings look at Earth from space with the Sun behind the viewer at the equinox, so the solar zenith angle at every point equals its latitude (equator at the center line, poles at top and bottom). The shell around the disk is the atmosphere's limb: the color of light scattered toward the viewer along a tangent ray at each altitude, computed with the same model. The atmosphere is drawn 30 times too thick, so 100 km spans half an Earth radius; a real limb is a hairline. The disk shows the planet's reflected color over a dark ocean (over ice for Snowball Earth), with limb brightness compressed so faint upper layers remain visible.

Modern, clean air. Modern Earth for reference: white-blue troposphere grading to a deep blue stratosphere and fading to black, brightest at the equator and darker toward the poles.

Early Archean, ~3.8 Ga. Clear early Archean. Structurally modern: white-blue near the surface, deep blue above, fading to black. With a little more CO₂ and no ozone the shell is marginally paler than today's and the disk very slightly whiter.

Neoproterozoic, ~2.7 Ga (thick haze). Hazy Neoproterozoic. The organic haze colors the entire limb tan: every tangent ray below ~60 km crosses the layer (centered ~45 km) twice. Only above it does the shell turn a faint, dusty blue. The disk is the pale orange dot, though from space it is a subtle cream-orange rather than a vivid one.

Early Hadean, ~4.4 Ga. 30-bar CO₂ Hadean. The shell is an opaque, luminous white-peach almost all the way up, because tangent rays at any tangent height below ~60 km pass through many optical depths of CO₂; only the outermost fringe thins to blue. The disk is bright and nearly white: a cloudless, Rayleigh-only Venus.

Snowball Earth, ~700 Ma. Snowball Earth. The bluest shell in the set above a blazing white disk of ice. The polar limb stays bright because the frozen surface reflects so much light back into the atmosphere.

Carboniferous, ~300 Ma. Carboniferous, 33% O₂. The extra oxygen raises total pressure by about a tenth, so the limb is marginally thicker and brighter than today's, but to the eye it is indistinguishable from modern Earth: once ozone and clean air are in place, the sky from space has been the same pale blue dot for two billion years.

K-Pg impact winter, 66 Ma. Impact winter. A dim brown-gray band from the soot and sulfate layer occupies the lower and middle shell; above ~40 km the untouched upper atmosphere is still faintly blue. The disk is a dull gray-lilac, and dark at high latitudes.

Volcanic year (Tambora 1815 / Toba-lite). Volcanic year. A milky white sulfate band near 20 km sits between the white-blue troposphere and the deep blue upper shell. The disk is brighter and whiter than clean modern Earth.

Epoch by epoch

The seventeen panels that follow are laid out identically so they can be compared at a glance. Each panel begins with the epoch's name and a one-line summary of the atmosphere assumed (Table 1 gives the full parameters). The upper row shows the noon sky dome at three latitudes—the equator, a mid-latitude site and the summer pole—with the zenith at the top of each swatch and the horizon at the bottom, the correlated color temperatures of both printed above, and the Sun's disk drawn in its own color, at its noon elevation, and with a brightness that reflects how much of it survives the atmosphere (where the Sun would not be visible at all, the swatch says so). The two strips beneath trace the sky from the solar horizon across the zenith to the antisolar horizon at two moments: with the Sun sitting on the horizon, and with it 4° below, in civil twilight. All swatches are shown at a brightness relative to today's clean sky, so a dim epoch reads as dim; the paragraph after each panel explains what the colors mean and why they arise.

Early Hadean, ~4.4 Ga

post-magma-ocean CO₂/steam atmosphere (~30 bar CO₂)

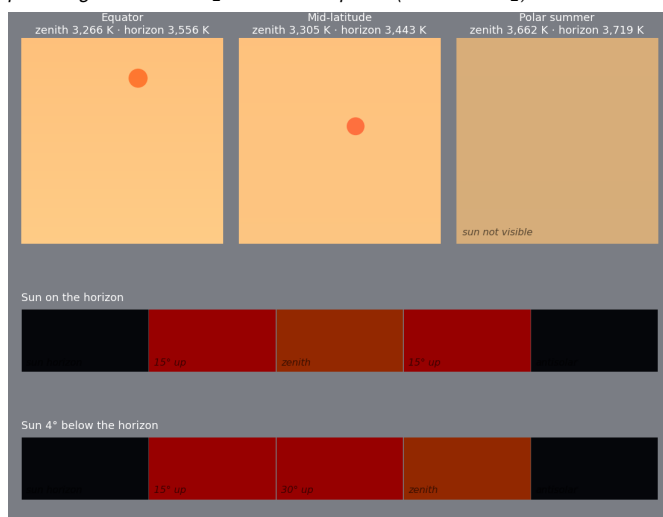


Figure 3: Early Hadean, ~4.4 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A ~30-bar CO₂ atmosphere has a Rayleigh optical depth near 5 at 550 nm and about 18 in the violet, so blue light is scattered back to space many times before it reaches the ground. What arrives is a diffuse, shadowless, peach-white glow of ~3,300 K, nearly identical from zenith to horizon, under a young Sun that is only 70% as bright. The Sun itself is a dim ember at the equator and vanishes entirely at mid and high latitudes: this is a Venus-like sky. Sunset is a brief, deep-orange dimming of the whole dome rather than an event at the horizon.

Late Hadean, ~4.0 Ga

CO₂ mostly drawn down into carbonate; ~1 bar N₂, ~0.5 bar CO₂

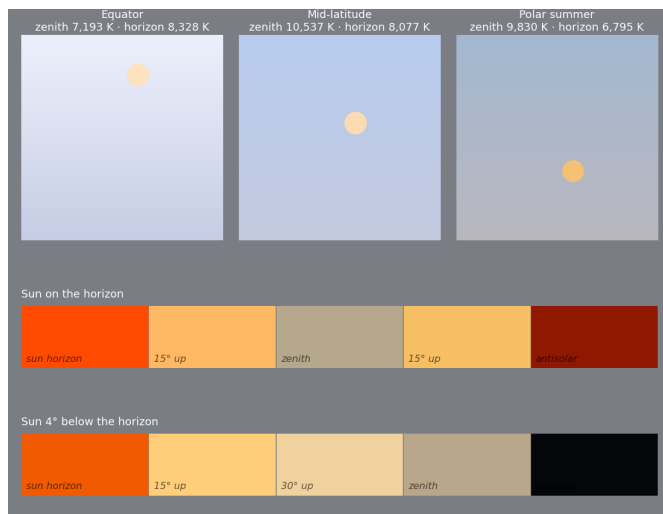


Figure 4: Late Hadean, ~4.0 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Once most of the CO₂ has gone into carbonate, Rayleigh scattering falls to roughly twice the modern value. The sky turns pale blue at the zenith but conspicuously whiter than today (7,000–10,500 K versus 9,000–15,000 K), because the extra scattering fills the dome with multiply-scattered white light. Horizons are milky. The setting Sun is deep red and the twilight zenith, with no ozone yet, is cream rather than blue.

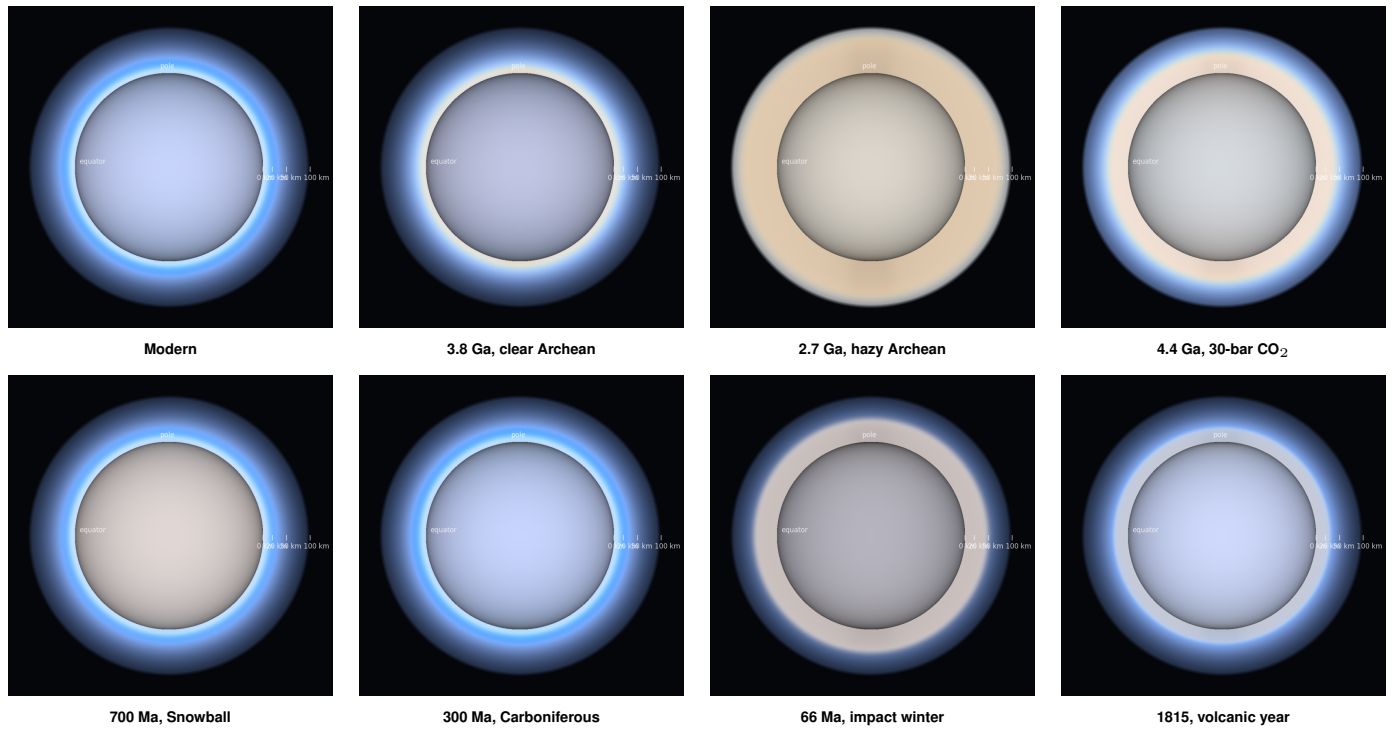


Figure 2: Volumetric renderings of the atmosphere at eight epochs. Equinox geometry with the Sun behind the viewer; the shell is the limb color along tangent rays, drawn 30 times too thick (100 km spans half an Earth radius); the disk is the planet's reflected color over ocean (ice for Snowball Earth). Limb brightness is compressed so faint upper layers remain visible.

Early Archean, ~3.8 Ga

haze-free, anoxic; $\text{CO}_2 \sim 0.1$ bar, $\text{CH}_4 \sim 1000$ ppm, no ozone

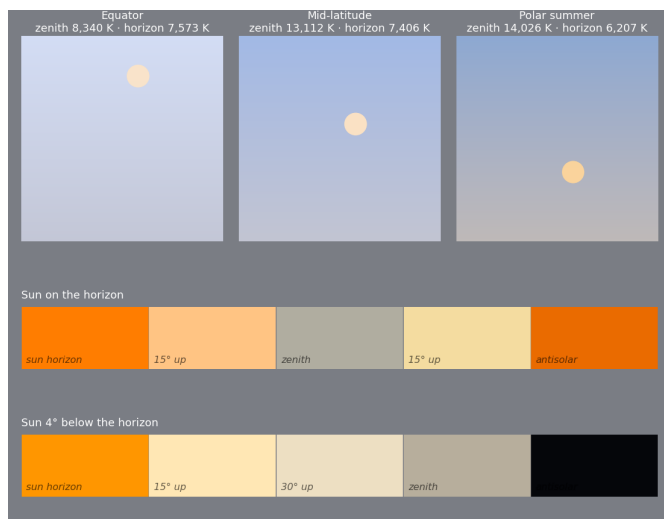


Figure 5: Early Archean, ~3.8 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

With CO_2 down to ~ 0.1 bar, noon skies are close to modern blue, only slightly whiter. The unmistakable difference is at dusk. The blue of the modern twilight zenith is produced by ozone absorbing yellow-orange light (the Chappuis band) along the long grazing path; with no free O_2 there is no ozone, so the Archean twilight zenith is a pale yellow-cream, and the whole sky fades from blue to beige to orange as the Sun sets. Sunsets themselves are much like today's, slightly redder from the extra CO_2 and dimmer Sun.

Neoproterozoic, ~2.7 Ga (thin haze)

biogenic $\text{CH}_4/\text{CO}_2 \sim 0.1$: intermittent thin organic haze

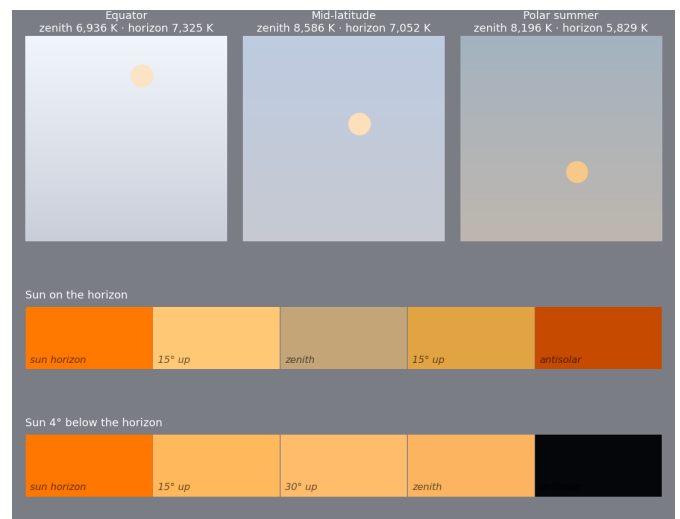


Figure 6: Neoproterozoic, ~2.7 Ga (thin haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

An intermittent organic haze at optical depth ~ 0.15 pushes the zenith toward a pastel, milk-blue (7,000–8,500 K) and lays a warm cast on the horizon; at the poles, where sunlight crosses the haze obliquely, the horizon turns visibly cream. Twilight is orange all the way to the zenith.

Neoproterozoic, ~2.7 Ga (thick haze)

"Pale Orange Dot": Titan-like fractal organic haze, $\tau_{550} \sim 0.6$

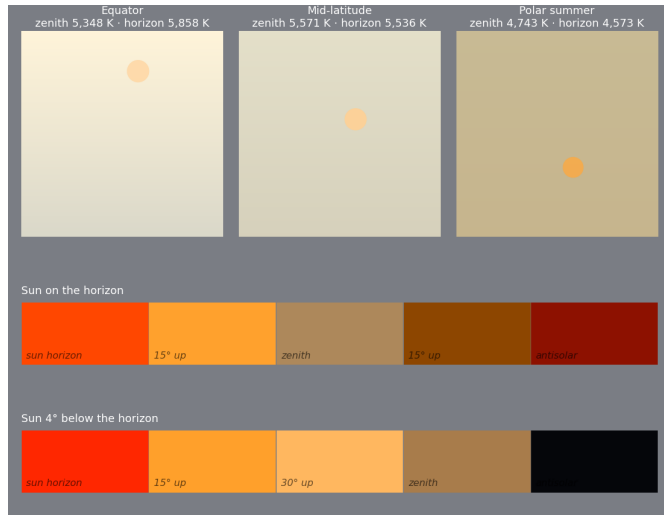


Figure 7: Neoproterozoic, ~2.7 Ga (thick haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

This is the 'Pale Orange Dot' seen from below. At optical depth ~ 0.6 the haze absorbs blue and scatters forward, so the entire dome is a warm cream to pale tan ($\sim 5,300$ K at the equator, $\sim 4,700$ K at the poles) with almost no zenith-to-horizon gradient: the sky looks like a bright overcast made of orange-tinted glass. The Sun is a soft, defined disk (haze scatters strongly forward) at 4,100 K. Sunsets are deep and long: the solar horizon reaches vermilion, the sky 15° above the Sun is amber, and the antisolar sky is peach rather than pink.

Neoproterozoic, ~2.7 Ga (very thick haze)

upper-end haze, $\tau_{550} \sim 1.5$: CH_4/CO_2 well above 0.2

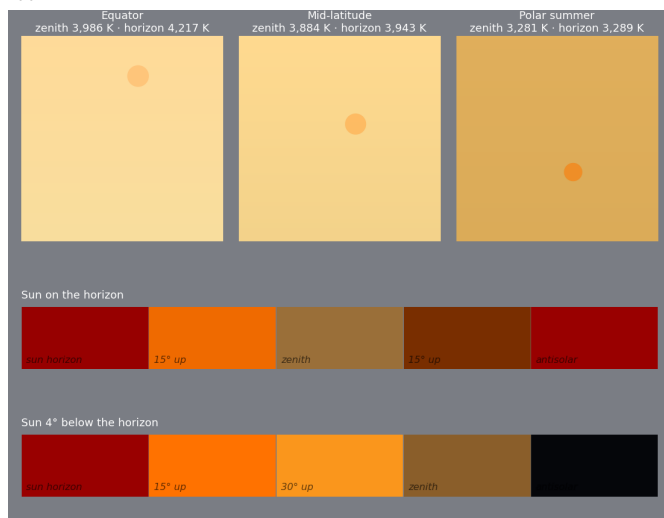


Figure 8: Neoproterozoic, ~2.7 Ga (very thick haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

At the upper end of plausible haze ($\tau \approx 1.5$) the sky is a uniform saturated apricot (3,300–4,200 K), fully Titan-like, and the polar Sun is reduced to an orange smear. Anything below this haze would have had roughly 20–30% of the surface sunlight of a clear sky.

Paleoproterozoic, ~2.2 Ga

after the Great Oxidation Event: $\text{O}_2 \sim 1\%$ PAL, ozone column 66 DU

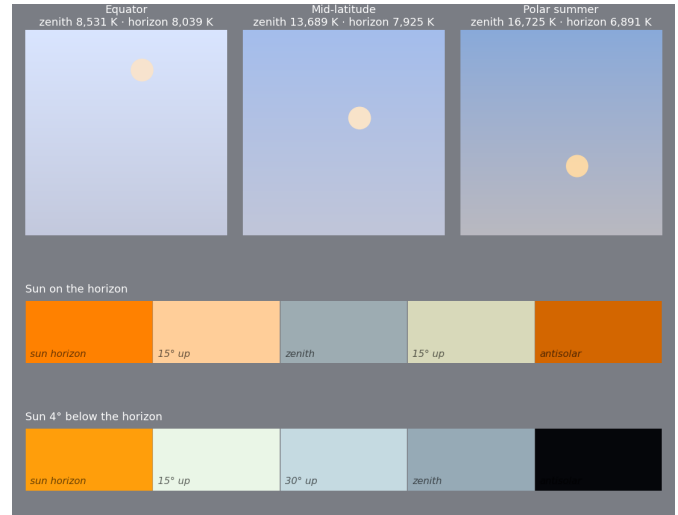


Figure 9: Paleoproterozoic, ~2.2 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

After the Great Oxidation Event, about 1% of today's O_2 builds an ozone column of 66 DU, the Cooke et al. (2021) global mean for that oxygen level: about a fifth of a modern mid-latitude column, not the half that older estimates gave. Noon skies are already modern blue. Twilight is where the thin column shows. With the Sun 4° below the horizon the zenith leaves the cream of an ozone-free dusk ($\sim 6,000$ K) and becomes a pale blue-gray ($\sim 7,100$ K). It is not yet the deep blue of a 300 DU twilight ($\sim 12,500$ K). That saturated blue dusk is in place once the column is near modern, as in the snowball sky at 169 DU. (The pink antisolar arch needs no ozone: it is a Rayleigh effect and shows up faintly even in the Hadean model.)

Snowball Earth, ~700 Ma

cold, dry, very clean air; ice/snow surface (albedo 0.75); ozone 169 DU

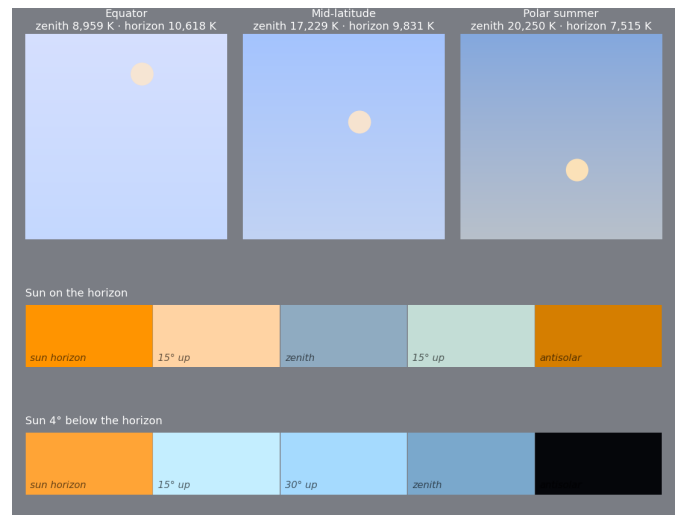


Figure 10: Snowball Earth, ~700 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A frozen planet has a dry, exceptionally clean atmosphere and a surface albedo near 0.75. The result is the bluest sky in Earth's history: 9,000 K at the equatorial zenith and over 20,000 K over polar ice, with horizons that stay distinctly blue instead of whitening, because the sunlit ice floods the sky from below with blue-white light. Sunsets are golden and comparatively pale; there is little aerosol to redden them.

Carboniferous, ~300 Ma

O_2 ~30-35%; total pressure ~1.1 bar; wildfire smoke common

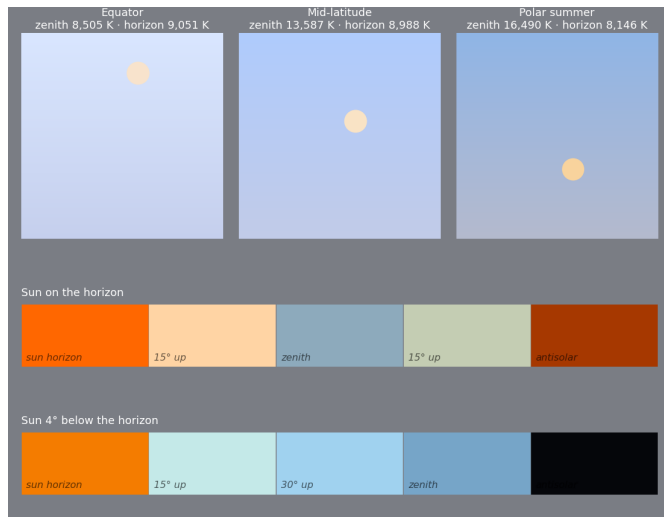


Figure 11: Carboniferous, ~300 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Thirty-plus percent O_2 raises the total pressure and Rayleigh scattering by ~10%, so skies are marginally brighter and horizons a touch whiter than today; the effect is subtle. Frequent wildfire smoke in the high- O_2 world would have produced red suns and brown-orange horizons far more often than in the modern era.

K-Pg impact winter, 66 Ma

months after Chicxulub: stratospheric soot ($\tau \sim 1.5$) + sulfate ($\tau \sim 0.5$)

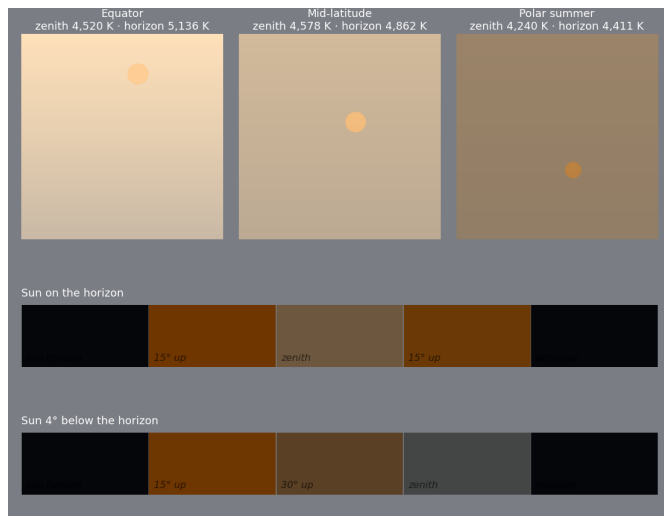


Figure 12: K-Pg impact winter, 66 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Months after the Chicxulub impact, a global stratospheric layer of soot and sulfate turns the sky into a dim, uniform amber-beige (4,200–5,100 K) with no blue at all, about one-fifth as bright as a clear sky at the equator and 20× dimmer at high latitudes. The Sun is a pale orange disk at low latitudes and effectively gone near the poles. Sunsets do not happen as color events: the whole dome simply fades to dark brown, and the twilight sky is gray.

ζ Ophiuchi supernova, ~1.78 Ma

clean Early Pleistocene air, under a magnitude –11.6 supernova in Scorpius

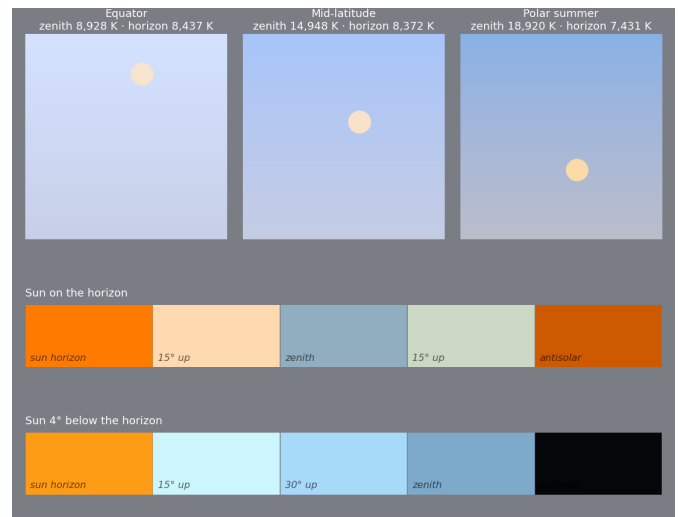


Figure 13: ζ Ophiuchi supernova, ~1.78 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

About 1.78 million years ago a star of 16 to 18 solar masses exploded in a binary in the Scorpius-Centaurus association. The explosion unbound its companion, which is now the runaway O star ζ Ophiuchi, and left a neutron star that is now the pulsar PSR B1706-16; tracing both back puts the explosion 1.78 ± 0.21 Myr ago at 107 ± 4 pc (Neuhäuser et al. 2019). Its iron-60 may be part of the 1.5 to 3.2 Myr old ^{60}Fe found in deep-sea crusts and on the Moon. A typical Type II-P supernova (Richardson et al. 2014) at that distance peaks near –11.6, about twice as bright as Geminga's, or –12.3 if the binary stripped the star and made it a Type Ib. The paper's abstract gives no sky position, so the star is drawn where a straight-line trace of ζ Oph's present motion puts it, near RA 16h10m, Dec –21°. The stars are moved back along their measured space motions too; Antares has since travelled about 12°, and in the sky of the time the supernova sat about 12° west of it. That trace lands at 130 to 140 pc rather than 107, so the position is approximate. From mid-northern latitudes it stays low in the south. The air is clean Early Pleistocene air.

Geminga supernova, ~342 ka

clean Middle Pleistocene air, under a magnitude –11 supernova in Orion

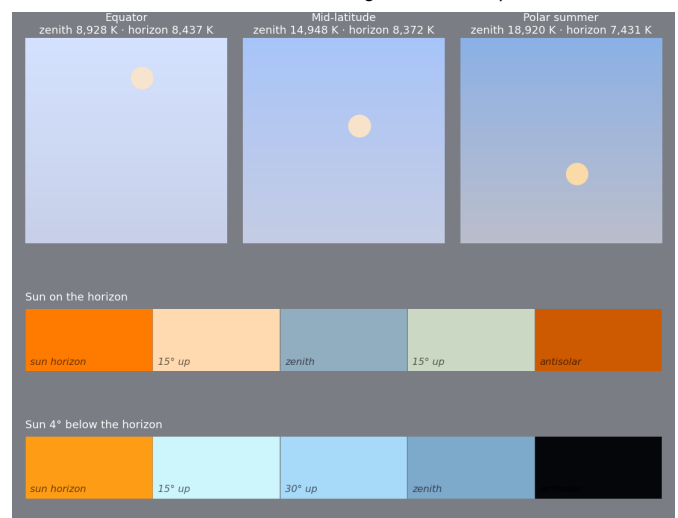


Figure 14: Geminga supernova, ~342 ka. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

About 342,000 years ago, going by the spin-down age of the pulsar it left behind (Salvati and Sacco 2008), a star of at most 15 solar masses exploded 90 to 240 pc away, in the direction of Orion (Pellizza et al. 2005). Type II-P supernovae peak near absolute magnitude

—16.75 (Richardson et al. 2014), so from the middle of that range, 147 pc, it reached about —11: a fifth of the full Moon's light, all from a point. The —13 often quoted needs both the near end of the distance range and an unusually bright explosion. It burned in the daytime sky as a dazzling star, cast sharp shadows at night, and held near peak for the three months of its plateau. The air is clean Middle Pleistocene air; at this distance the explosion barely touches the atmosphere (Thomas et al. 2016), so the sky's colors are today's. The star is drawn at the birthplace. The stars around it are moved back along their measured space motions, which puts it just 1.4° from Betelgeuse in the sky of the time.

Volcanic year (Tambora 1815 / Toba-lite)

stratospheric sulfate veil, $\tau_{550} \sim 0.4$

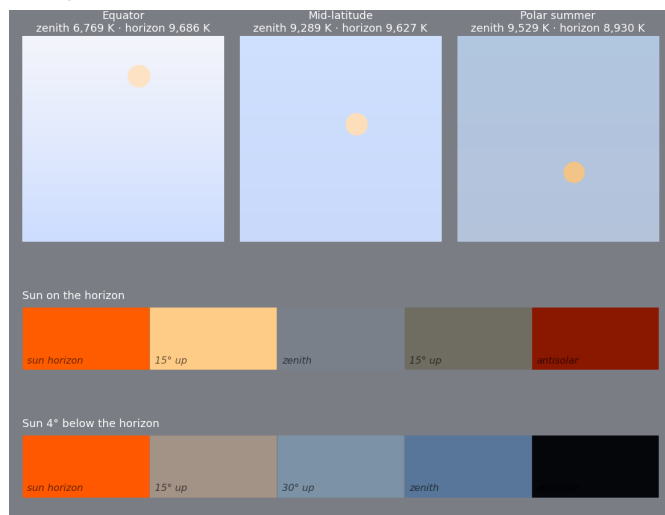


Figure 15: Volcanic year (Tambora 1815 / Toba-lite). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A Tambora-scale sulfate veil ($\tau \approx 0.4$) is what most people picture as a spectacular sky, and the model agrees. Noon skies are milky, whiter at the zenith than the horizon (the reverse of a clean sky). Sunsets glow salmon-orange, and the after-sunset sky $15\text{--}30^\circ$ above the horizon turns a luminous pink-lavender: the high sulfate layer is still lit when the lower atmosphere is in shadow. This is the sky of Turner's 1816 paintings and of the Krakatoa twilights of 1883.

Ozone-hole spring

Antarctic spring in the ozone-hole years: column 130 DU

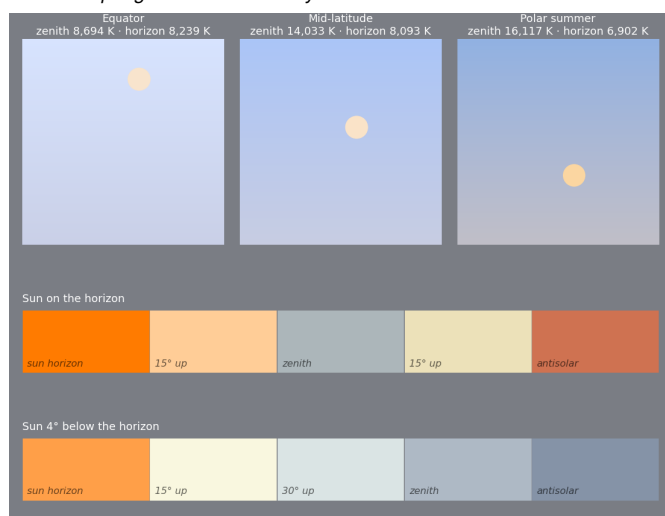


Figure 16: Ozone-hole spring. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Antarctic spring in the ozone-hole years, with the column cut to 130 DU and the layer lowered toward 18 km. Noon stays blue. Twilight is where the hole shows: with the Sun 4° down the zenith is a pale blue

($\sim 7,800$ K) instead of the deep blue ($\sim 12,500$ K) of a 300 DU sky. It is the same direction as the thin post-oxidation column, and still well short of the cream ($\sim 6,000$ K) of a sky with no ozone at all.

Modern, clean air

aerosol optical depth 0.1; ozone 260/300/330 DU by latitude

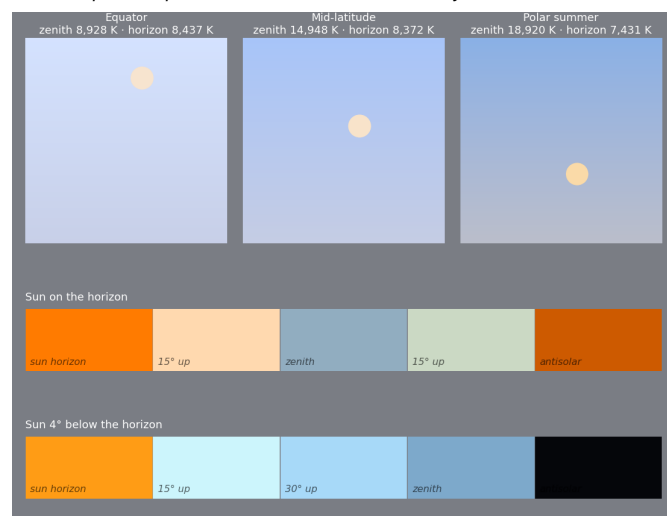


Figure 17: Modern, clean air. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Baseline. Equatorial zenith $\sim 9,000$ K; mid-latitude $\sim 15,000$ K; polar summer $\sim 19,000$ K. Horizons run 7,400–8,400 K, essentially white. Setting Sun $\sim 1,700$ K; solar horizon orange; antisolar horizon pink.

Modern, polluted megacity

aerosol optical depth 0.6, slightly absorbing

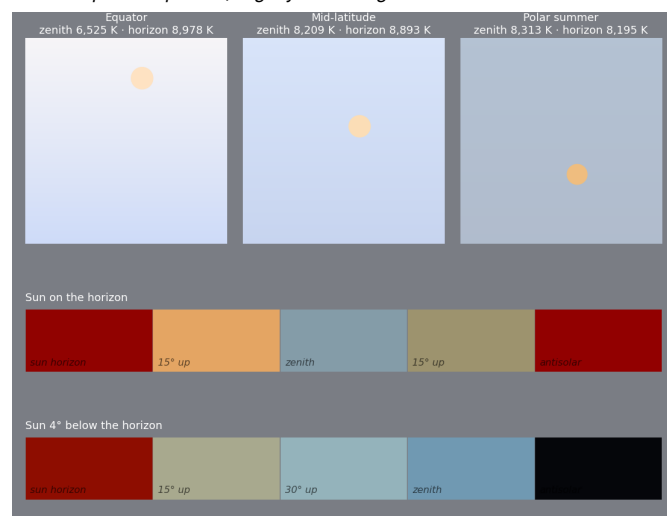


Figure 18: Modern, polluted megacity. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Aerosol optical depth 0.6 flattens the sky to a near-uniform pale gray-blue (6,500–8,300 K), brighter and whiter at the zenith than a clean sky, with the horizon slightly bluer than the zenith. The Sun disappears into gray murk well before it reaches the horizon.

Year 2100

today's clean air, with the filed megaconstellation and the Sunrise orbital datacenters

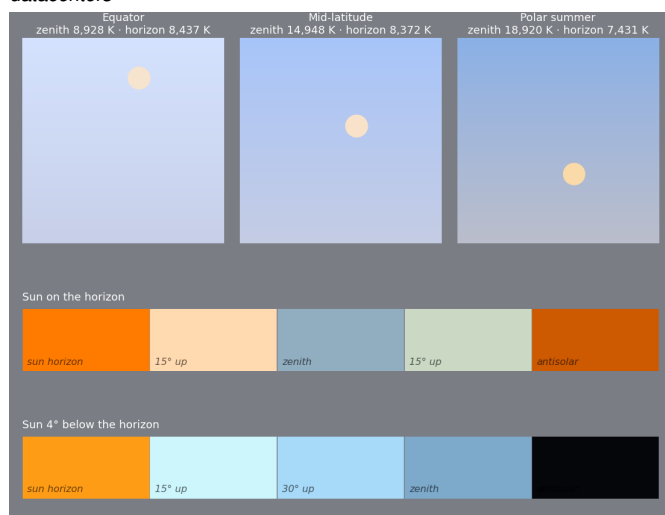


Figure 19: Year 2100. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

The air is today's. What changes is the traffic. Lawler, Boley, and Rein (2022) put every filed megaconstellation on orbit, about 65,000 satellites, and found the worst naked-eye light pollution near 50° latitude. Each of those is their diffuse sphere, effective area 0.8 m², so its magnitude is on the same scale as the stars. At the equinox a mid-latitude sky has a few hundred of them above naked-eye brightness in the hour after sunset, and Earth's shadow takes them by midnight. On top of that fleet, Boley, Lawler, and Rein (2026) model orbital datacenters. Of the three filed designs, this sky takes the middle one and the more optimistic of their two node spreads: Blue Origin's Sunrise, 51,600 Sun-synchronous satellites from 500 to 1,800 km, in a dawn-dusk cross of rings whose nodes wander 10° to either side of the terminator. Each is a diffuse sphere of 800 m² at albedo 0.2, the size the paper calls conservative next to panels several times larger, and two hundred times the reflecting area of a communications satellite. An hour after sunset at mid-latitudes, thousands of them are above naked-eye brightness, a bright band along the terminator. The paper's all-night winter rings belong to the December solstice, which this equinox clock does not show. A low orbit crosses the dome in minutes.

Stars brighter than Sirius

Sirius, at magnitude −1.46, is the brightest star in today's night sky, but it has not always been. The Sun's neighbours drift past it at tens of kilometres a second, so a star a hundred parsecs away now may have passed within a few parsecs of it a few million years ago. Tracing every Hipparcos star that has a radial velocity back 10 Myr through the Galaxy, with 2,000 draws over each star's measurement errors, finds eleven that outshone today's Sirius (Table 3). The brightest is θ Columbae, a fifth-magnitude star today, which passed about 2 pc from the Sun 4.7 million years ago and peaked near magnitude −5, as bright as Venus at its best; Gaia DR3 astrometry gives the same (−5.2 at 2.1 pc, 5.1 million years ago). ϵ Canis Majoris reached −4.35 4.4 million years ago, a result known since Tomkin (1998), and β Canis Majoris −3.6 at the same time. From about 3.5 to 5 million years ago the night sky held at least two stars brighter than any star today, and at times three.

These stars now lie in Columba, Canis Major and Puppis because that is the solar antapex: the Sun moves toward Hercules at about 20 km/s relative to its neighbours, so the stars it has passed end up behind it. The closest pass, ζ Leporis at 1.3 pc 850,000 years ago, came as close as α Centauri is now, far too distant to disturb the planets. Only distance changes the brightness here. ϵ and β Canis

Majoris are giants 12 to 22 million years old and were probably a few tenths of a magnitude fainter then. Algol's radial velocity (3.7 ± 3.9 km/s, hard to measure in an eclipsing binary) is too uncertain to say whether it passed within 1.3 or 28 pc of the Sun. Rigel, Saiph, Mintaka, Alnitak and ξ Persei also come out bright, but 6 to 10 million years ago, about their own ages, so they are left out. The site's skies do not show these peaks: the supernova epochs (1.78 Ma and 342 ka) fall between them, though at 342 ka Aldebaran is close to its peak of −1.5.

Caveats

Paleoatmospheric compositions carry order-of-magnitude uncertainty, and the haze optical properties are taken from Titan-analog tholins; the three haze cases bracket that uncertainty. Aerosol loads for every pre-Cenozoic epoch are educated guesses. The color-matching functions use an analytic fit, and colors are shown without chromatic adaptation, so the tints represent what a modern camera set to day-light balance would record rather than what an adapted observer would "see" (an adapted eye would perceive the hazy Archean sky as nearer to white and the modern sky as bluer than shown). Clouds are omitted throughout. The two-stream multiple-scattering term is approximate for the thickest atmospheres, where a full Monte Carlo treatment would refine the exact tint of the Hadean sky.

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Table 3: Stars that outshone today's Sirius ($V = -1.46$) in the last 10 million years, traced back through the Galaxy from Hipparcos astrometry (XHIP) with 2,000 draws over the measurement errors. Peak magnitudes are medians with the 68% range; the last column is the fraction of draws brighter than Sirius today. Brightness changes with distance only.

Star	Type	Today: V , pc	Peak V (68% range)	Closest (pc)	Myr ago	Brighter than Sirius
θ Columbae	B8:IV	5.00, 221	-5.00 (-5.77 to -4.41)	2.2	4.73	100%
ε Canis Majoris (Adara)	B1III	1.50, 124	-4.35 (-4.44 to -4.26)	8.4	4.42	100%
β Canis Majoris (Mirzam)	B1III	1.98, 151	-3.58 (-3.72 to -3.45)	11.7	4.35	100%
ρ Orionis	K0 III	4.46, 105	-3.16 (-4.24 to -2.40)	3.1	2.54	100%
Algol	B7V	2.09, 28	-2.88 (-4.65 to 2.09)	2.9	5.07	62%
ζ Leporis	A2IV-V(n)	3.55, 22	-2.54 (-2.63 to -2.43)	1.3	0.85	100%
ν Puppis	B7III	3.17, 114	-2.42 (-2.64 to -2.22)	8.7	3.56	100%
ζ Sagittarii (Ascella)	A2.5Va	2.60, 27	-1.97 (-2.07 to -1.87)	3.3	1.05	100%
Canopus	A9II	-0.62, 95	-1.92 (-2.01 to -1.82)	52.2	3.18	100%
Aldebaran	K5III	0.87, 20	-1.50 (-1.53 to -1.47)	6.8	0.33	92%
ζ Canis Majoris (Furud)	B2.5 V	3.02, 111	-1.47 (-1.87 to -0.94)	14.0	3.27	51%

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Appendix: computed chromaticities

Table 4: CIE 1931 chromaticity (x, y) and correlated color temperature of the noon zenith and horizon sky, and direct-Sun brightness relative to the modern equatorial Sun.

Epoch	Latitude	Zenith (x, y)	CCT (K)	Horizon (x, y)	CCT (K)	Sun
Early Hadean, ~4.4 Ga	Equator	0.421, 0.400	3,266	0.406, 0.400	3,556	0.0086
Early Hadean, ~4.4 Ga	Mid-latitude	0.418, 0.399	3,305	0.412, 0.400	3,443	0.0021
Early Hadean, ~4.4 Ga	Polar summer	0.399, 0.391	3,662	0.396, 0.392	3,719	1.4e-06
Late Hadean, ~4.0 Ga	Equator	0.303, 0.317	7,193	0.292, 0.302	8,328	0.63
Late Hadean, ~4.0 Ga	Mid-latitude	0.275, 0.288	10,537	0.294, 0.305	8,077	0.55
Late Hadean, ~4.0 Ga	Polar summer	0.277, 0.296	9,830	0.309, 0.320	6,795	0.25
Early Archean, ~3.8 Ga	Equator	0.291, 0.303	8,340	0.299, 0.310	7,573	0.77
Early Archean, ~3.8 Ga	Mid-latitude	0.264, 0.274	13,112	0.301, 0.313	7,406	0.72
Early Archean, ~3.8 Ga	Polar summer	0.259, 0.272	14,026	0.318, 0.330	6,207	0.47
Neoproterozoic, ~2.7 Ga (thin haze)	Equator	0.306, 0.323	6,936	0.301, 0.316	7,325	0.72
Neoproterozoic, ~2.7 Ga (thin haze)	Mid-latitude	0.288, 0.304	8,586	0.305, 0.320	7,052	0.64
Neoproterozoic, ~2.7 Ga (thin haze)	Polar summer	0.290, 0.312	8,196	0.325, 0.340	5,829	0.31
Neoproterozoic, ~2.7 Ga (thick haze)	Equator	0.336, 0.355	5,348	0.324, 0.346	5,858	0.46
Neoproterozoic, ~2.7 Ga (thick haze)	Mid-latitude	0.331, 0.353	5,571	0.332, 0.353	5,536	0.35
Neoproterozoic, ~2.7 Ga (thick haze)	Polar summer	0.355, 0.377	4,743	0.361, 0.379	4,573	0.07
Neoproterozoic, ~2.7 Ga (very thick haze)	Equator	0.386, 0.396	3,986	0.377, 0.396	4,217	0.19
Neoproterozoic, ~2.7 Ga (very thick haze)	Mid-latitude	0.393, 0.405	3,884	0.391, 0.406	3,943	0.1
Neoproterozoic, ~2.7 Ga (very thick haze)	Polar summer	0.429, 0.422	3,281	0.429, 0.423	3,289	0.004
Paleoproterozoic, ~2.2 Ga	Equator	0.290, 0.303	8,424	0.295, 0.306	7,928	0.87
Paleoproterozoic, ~2.2 Ga	Mid-latitude	0.264, 0.273	13,288	0.297, 0.308	7,780	0.81
Paleoproterozoic, ~2.2 Ga	Polar summer	0.255, 0.267	15,479	0.311, 0.323	6,643	0.51
Snowball Earth, ~700 Ma	Equator	0.289, 0.297	8,737	0.278, 0.288	10,229	0.99
Snowball Earth, ~700 Ma	Mid-latitude	0.256, 0.262	16,181	0.282, 0.294	9,498	0.93
Snowball Earth, ~700 Ma	Polar summer	0.249, 0.259	18,509	0.302, 0.319	7,234	0.62
Carboniferous, ~300 Ma	Equator	0.290, 0.303	8,421	0.287, 0.296	8,937	0.91
Carboniferous, ~300 Ma	Mid-latitude	0.262, 0.272	13,512	0.287, 0.296	8,930	0.82
Carboniferous, ~300 Ma	Polar summer	0.252, 0.264	16,735	0.293, 0.302	8,178	0.41
K-Pg impact winter, 66 Ma	Equator	0.362, 0.369	4,496	0.343, 0.353	5,096	0.12
K-Pg impact winter, 66 Ma	Mid-latitude	0.360, 0.367	4,558	0.350, 0.360	4,839	0.054
K-Pg impact winter, 66 Ma	Polar summer	0.372, 0.374	4,223	0.365, 0.369	4,401	0.00038
ζ Ophiuchi supernova, ~1.78 Ma	Equator	0.287, 0.299	8,852	0.291, 0.301	8,414	1
ζ Ophiuchi supernova, ~1.78 Ma	Mid-latitude	0.259, 0.267	14,903	0.291, 0.301	8,411	0.92
ζ Ophiuchi supernova, ~1.78 Ma	Polar summer	0.247, 0.257	19,331	0.299, 0.310	7,549	0.53
Geminga supernova, ~342 ka	Equator	0.287, 0.299	8,852	0.291, 0.301	8,414	1
Geminga supernova, ~342 ka	Mid-latitude	0.259, 0.267	14,903	0.291, 0.301	8,411	0.92
Geminga supernova, ~342 ka	Polar summer	0.247, 0.257	19,331	0.299, 0.310	7,549	0.53
Volcanic year (Tambora 1815 / Toba-lite)	Equator	0.310, 0.324	6,720	0.282, 0.293	9,561	0.67
Volcanic year (Tambora 1815 / Toba-lite)	Mid-latitude	0.283, 0.296	9,242	0.281, 0.293	9,587	0.53
Volcanic year (Tambora 1815 / Toba-lite)	Polar summer	0.280, 0.296	9,537	0.285, 0.299	8,997	0.13
Ozone-hole spring	Equator	0.288, 0.300	8,694	0.293, 0.303	8,239	1
Ozone-hole spring	Mid-latitude	0.261, 0.270	14,033	0.294, 0.304	8,093	0.94
Ozone-hole spring	Polar summer	0.254, 0.265	16,117	0.307, 0.319	6,902	0.57
Modern, clean air	Equator	0.287, 0.299	8,852	0.291, 0.301	8,414	1
Modern, clean air	Mid-latitude	0.259, 0.267	14,903	0.291, 0.301	8,411	0.92
Modern, clean air	Polar summer	0.247, 0.257	19,331	0.299, 0.310	7,549	0.53
Modern, polluted megacity	Equator	0.313, 0.328	6,478	0.287, 0.298	8,877	0.6
Modern, polluted megacity	Mid-latitude	0.292, 0.306	8,170	0.287, 0.299	8,854	0.45
Modern, polluted megacity	Polar summer	0.290, 0.306	8,331	0.291, 0.306	8,227	0.08
Year 2100	Equator	0.287, 0.299	8,852	0.291, 0.301	8,414	1
Year 2100	Mid-latitude	0.259, 0.267	14,903	0.291, 0.301	8,411	0.92
Year 2100	Polar summer	0.247, 0.257	19,331	0.299, 0.310	7,549	0.53